Acoustic Cues and Cross-Linguistic Perception of Checked Tones in Southern Min

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Introduction Checked tones in Southern Min, namely Tone 3 (Low) and Tone 5 (High), occur with unreleased codas /k, p, t, ʔ/. Previous studies show that acoustic cues distinguish Tones 3 and 5 [1], and that F1 and F2 offsets are crucial for identifying unreleased stops in Korean and Thai [2]. However, acoustic cues of checked-tone codas remain underexplored. These codas are often produced as damped or weakened signals [1], and tonal contrasts rely on multidimensional acoustic cues [3], potentially increasing perceptual difficulty. While perception studies on unreleased stops in Thai reported high perceptual accuracy for both Thai and English speakers (Thai speakers showing a slight advantage) [4], perceptual evidence for Southern Min checked tones remains limited. To address these gaps, this study focuses on how checked-tone codas in Southern Min contrast acoustically and perceptually across language backgrounds.

Method Four male Southern Min speakers were recorded producing eight targets (/ak, ap, at, aʔ/ with two tones), yielding 24 tokens each. Vowel and Coda were annotated in Praat. Acoustic parameters were extracted per ms in VoiceSauce. These acoustic measures were analyzed with linear mixed-effects models and likelihood ratio tests on time-normalized trajectories, separately in the coarticulation (40–60%, vowel–coda transition) and coda (60–100%, coda-only) windows. An online perception experiment via GitHub was conducted using stimuli from the four speakers with 65 participants across five L1 groups (Mandarin, English, Japanese, Korean, Southern Min), selected to leverage cross-linguistic differences in cue sensitivity to examine coda contrasts. Participants completed multiple-choice tone (16 trials) and coda (24 trials) identification tasks, with practice before each task. Results were analyzed using ANOVA implemented in Python.

Results & Discussion [*Production*] In Fig. 1: in the coarticulation window, /p/ exhibits significantly lower F1 than other codas ($p < .01$), while /t/ shows significantly higher F2 ($p < .05$); in the coda window, /k/ shows significantly higher F3 ($p < .05$). F0 differs significantly between Tones 3 and 5 ($p < .001$) and most coda pairs ($p < .05$). Energy trajectories show a significant coda contrast ($p < .05$), with faster energy decay for /p/ and /ʔ/ than for /k/ and /t/. Vowel duration consistently distinguishes /ʔ/, with significantly longer duration, from other codas ($p < .001$). Cepstral Peak Prominence differs significantly between tones ($p < .001$) and most coda pairs ($p < .05$). /t/ shows significantly lower H1*-A3* values in tone 5 ($p < .05$). Overall, /ʔ/ is the most acoustically distinct coda, while /k/ is the least distinct. **[*Perception*]** In Fig. 2, Mandarin listeners show the highest tone identification accuracy, while Southern Min listeners show lower accuracy than other groups ($p < .05$). Tone 3 (Low) is identified more accurately than Tone 5 (High). Coda identification is highest among Korean listeners, followed by English listeners; both groups exceed Japanese listeners ($p < .05$) and the remaining groups ($p < .001$). For /t/, Korean and English listeners show higher accuracy than other groups ($p < .05$), reflecting sensitivity to F2-related cues, while higher /k/ accuracy for Korean ($p < .05$) and Japanese may relate to sensitivity to F3-related cues [5]. Across groups, coda confusions mainly involve /k, p, t/, with /k/ showing lower accuracy, while /ʔ/ remains highly identifiable. While most groups' results align with expectations based on whether unreleased codas or close analogues (e.g., Japanese Sokuon) are present in their L1, Southern Min listeners show lower accuracy, potentially because the weakening of codas and their phonological characteristics leads them to rely less on segmental cues. In conclusion, checked-tone codas show systematic acoustic contrasts, and cross-linguistic perception reflects these contrasts while revealing an unusual pattern: these codas are acoustically contrastive but not robustly perceived by native speakers.

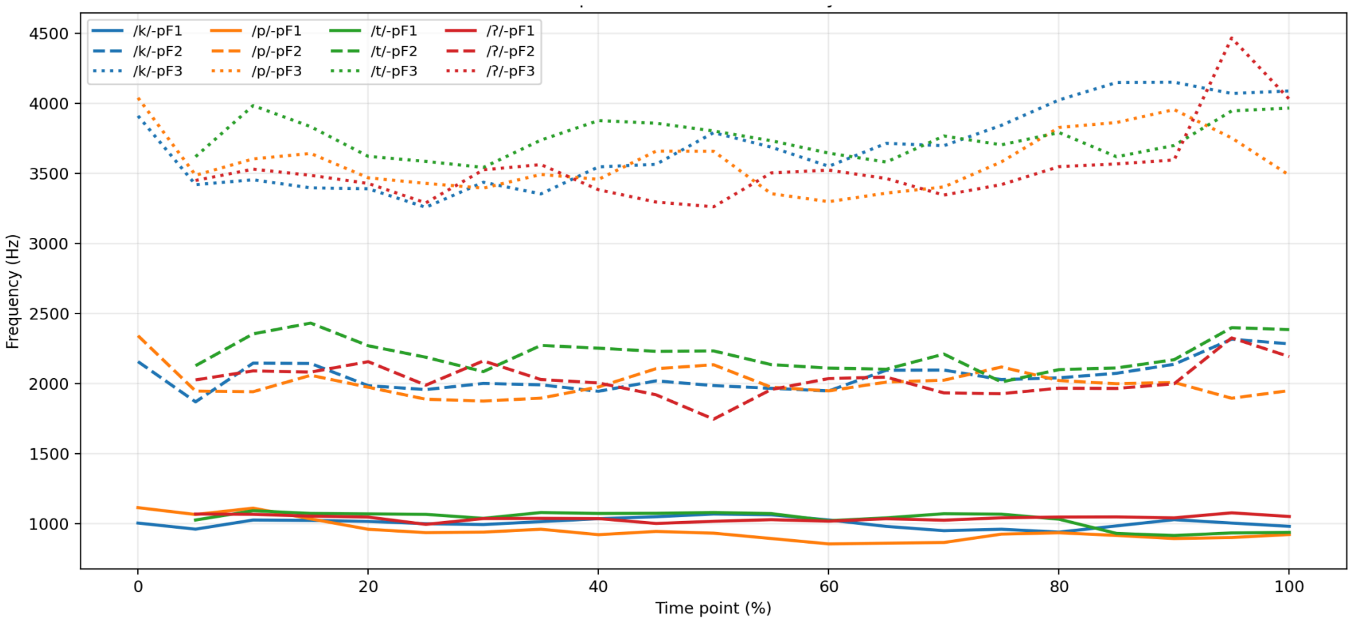

Fig. 1. Mean time-normalized trajectories of F1–F3 for the four codas, averaged across speakers.

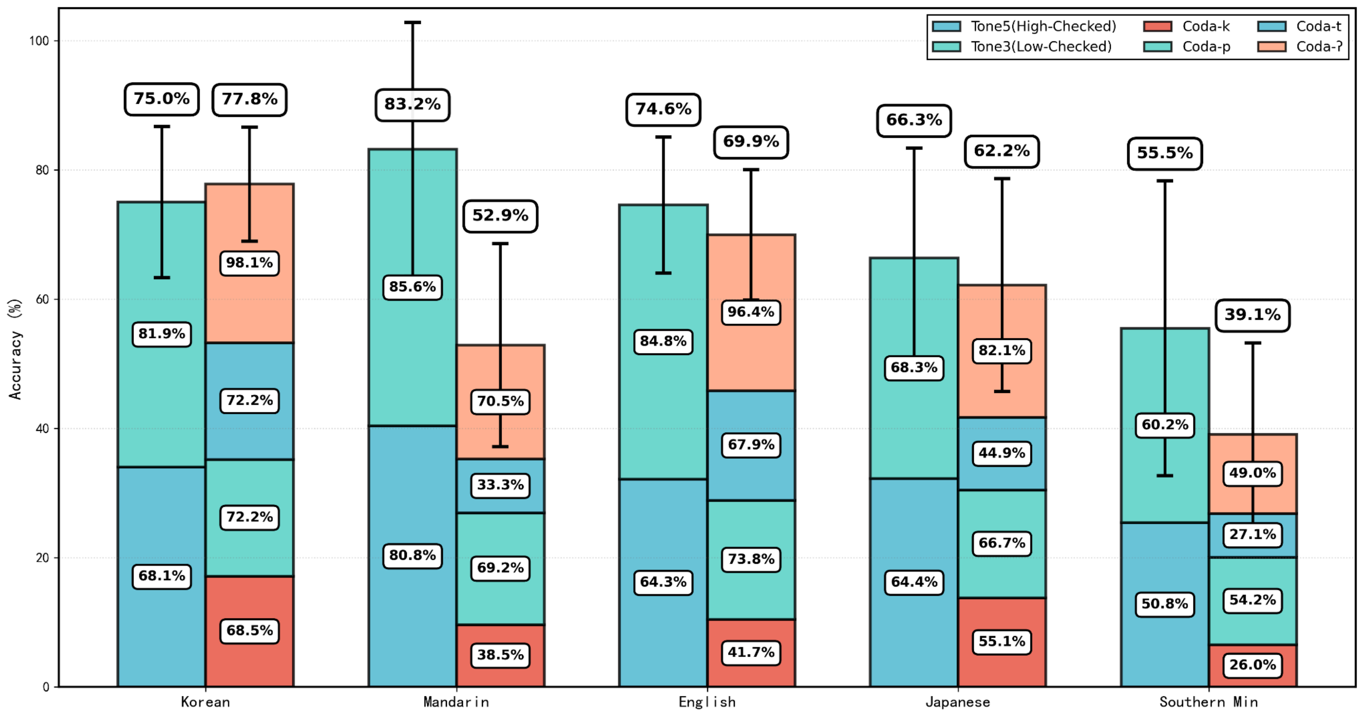

Fig. 2. Mean tone and coda identification accuracy across language groups with error bars, and stacked bars show the relative accuracy of individual tones and coda types within each group.

References [1] Pan, H. H. (2017). Glottalization of Taiwan Min checked tones. *Journal of the International Phonetic Association*, 47(1), 37-63. **[2]** Tsukada, K. (2004). A cross-linguistic acoustic comparison of unreleased word-final stops: Korean and Thai. In *INTERSPEECH* (pp. 1293-1296). **[3]** Huang, Y. (2024). Same F0, Different Tones: A Multidimensional Investigation of Southern Min Tones. In *INTERSPEECH* (pp. 4728-4732). **[4]** Abramson, A. S., & Tingsabadh, K. (1999). Thai final stops: Cross-language perception. *Phonetica*, 56(3-4), pp.111-122. **[5]** Iverson, P., Kuhl, P. K., Akahane-Yamada, R., Diesch, E., Tohkura, Y. I., Kettermann, A., & Siebert, C. (2003). A perceptual interference account of acquisition difficulties for non-native phonemes. *Cognition*, 87(1), pp.B47-B57.